The step by step execution of power (2, 5) is shown in Table 8.1. Each row shows the values of the two variables p and i at the end of an iteration, and how they are calculated. We see that $p = a^i$ is true at the start of the loop, and remains true in each row. Therefore, it is a loop invariant.

iteration	p	p x a	i	i+1	a^i
0	1		0		2^0
1	2	1x2	1	0 + 1	2^1
2	4	2x2	2	1 + 1	2^2
3	8	4x2	3	2 + 1	2^3
4	16	8x2	4	3 + 1	2^4
5	32	16x2	5	4+1	2^5

Table 8.1: Trace of power (2, 5)

When the loop ends, $p = a^1$ is still true, but i = 5. Therefore, $p = a^5$. In general, when the loop ends, $p = a^n$. Thus, we have verified that power(a, n) satisfies its specification.

Example 8.6. Recall the Chocolate bar problem of Example 6.11. How many cuts are needed to break the bar into its individual squares?

We decided to represent the number of pieces and the number of cuts by variables p and c respectively. Whenever a cut is made, the number of cuts increases by one and the number of pieces also increases by one. We decided to model it by an assignment.

p, c := p + 1, c+1

The process of cutting the bar can be modeled by a loop. We start with one piece and zero cuts, p = 1 and c = 0. Let n be the number of individual squares. When the number of pieces p equals the number of individual squares n, the process ends.

p, c : = 1 , 0

while p ≠ n

p, c := p + 1, c+1

We have observed (in Example 8.2) that p - c is an invariant of the assignment p, c := p + 1, c + 1. Let p - c = k, where k is a constant. The points in the algorithm where p - c = k is true are shown in the algorithm below, and in the flowchart of Figure 8.2.

p, c : = 1 , 0

1. -- p - c = k

while p ≠ n

2. -- p - c = k

p, c := p+1, c+1

3. -- p - c = k

4. --p-c=k,p=n

The loop invariant p- c = k is True at the start of the loop (line 1). Moreover, at the start of the loop, p- c = 1. Therefore, k = 1, and the loop invariant is p - c = 1

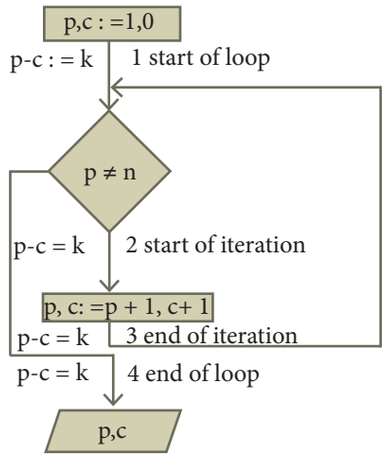


Figure 8.2: The points where the loop invariant is true

When the loop ends (line 4), the loop invariant is still true (p - c = 1). Moreover, the loop condition is false (p = n). From p - c = 1 and p = n,

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